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generative AI-assisted learning, metacognitive calibration, and structured reflection in adolescents

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INTRODUCTION

Learning is more than knowing the right answer—it also involves knowing how well we know it. As generative AI becomes increasingly embedded in learning, students can access explanations and support with unprecedented ease, but this raises an important question: does AI help learners become more accurate judges of their own knowledge? This question is particularly important during adolescence, when metacognitive skills are still developing. The present study explores how different forms of AI-supported learning, with and without structured reflection, shape adolescents' ability to accurately assess their knowledge both immediately and over time.

1. RESEARCH PROBLEM:

Neal Stephenson's THE DIAMOND AGE presents the "Young Lady's Illustrated Primer," a fictional educational technology that adapts to a learner's interests, pace, and developmental needs [1]. Today, Generative Artificial Intelligence (GenAI) is bringing increasingly interactive forms of educational assistance into real learning environments. While GenAI can support learners through explanations, examples, and feedback, successful task performance with AI assistance may not necessarily reflect independent understanding.

This distinction makes metacognitive calibration—the correspondence between a learner's confidence and objective performance—important. Existing research shows that GenAI can improve task performance while metacognitive accuracy and self-assessment do not necessarily improve correspondingly [2], [4], [5], [8]. This issue is particularly relevant for adolescents, as experimental research remains more heavily concentrated on university populations, despite emerging evidence involving secondary and upper-secondary learners [6], [8], [10], [11].

Therefore, there is a need to examine whether GenAI-assisted learning affects adolescents' confidence–performance calibration, and whether this relationship persists beyond immediate learning. Investigating calibration at immediate and delayed intervals can provide insight into an aspect of AI-assisted learning that conventional performance measures alone may not capture. This study therefore proposes structured metacognitive reflection as a potential means of addressing inaccurate confidence following AI-assisted learning, by prompting learners to evaluate uncertainty, verify AI-generated information, and assess their independent understanding.

2. EXISTING LITERATURE:

(2.1) GenAI-Assisted Learning: Performance, Cognitive Offloading, and Confidence

Research indicates that generative AI can improve reasoning, problem-solving, and academic performance across educational contexts [2], [4], [5], [6]. However, improved AI-assisted performance does not necessarily reflect independent learning, accurate understanding, or lasting retention. GenAI may also encourage cognitive offloading and reliance while affecting learners' confidence and self-assessment [7], [12]. Studies show that LLM assistance can enhance reasoning performance without resolving self-assessment difficulties [2]. Research on students' integration of ChatGPT-generated content into academic writing further highlights the role of epistemic beliefs and metacognitive accuracy in evaluating AI-generated information [9]. Evidence from a randomized study of AI-supported studying also raises questions about whether improved assisted performance necessarily translates into stronger unaided retention [16]. Thus, performance, confidence, and independent learning may not develop in parallel, underscoring the need to evaluate GenAI beyond immediate task performance.

(2.2) Metacognition, Calibration, and Reflective Engagement with GenAI

Distinguishing actual performance from perceived performance is central to metacognition, which involves regulating and understanding one's cognitive processes. Research on GenAI has examined metacognitive awareness, self-assessment, and regulation [10], [13], [14]. Findings show that LLM-generated feedback does not consistently improve self-assessment accuracy [8], and that improved AI-assisted performance may coexist with inaccurate judgments of performance [2]. Building on this, recent research suggests that metacognitive reflection can support more critical engagement with AI outputs and calibration by prompting learners to question AI outputs, compare them with their own reasoning or external evidence, and revise their judgments when discrepancies arise [7], [11], [13]. Thus, the literature increasingly suggests that GenAI's cognitive impact depends not only on the assistance provided, but also on learners' ability to reflect on AI outputs and adjust their confidence accordingly.

(2.3) Adolescent Learners and Research Gap

As GenAI research in education is expanding, experimental evidence remains concentrated in higher education [5], [6]. Emerging studies with secondary and upper-secondary learners examine self-assessment, metacognition, and trust in AI outputs [8], [10], [11], [15], but remain limited and heterogeneous in their designs and measures. Yet few studies isolate item-level confidence–accuracy calibration in controlled learning tasks, particularly among adolescents [2], [3], [4], [8], [10], [11], leaving a clear gap in understanding how AI-assisted performance relates to students' perceived

knowledge.

Although existing research has established that generative AI can influence learning performance, confidence, and metacognitive monitoring, these outcomes have largely been investigated separately or in different task contexts [2], [4], [5], [7]. Recent evidence has begun to examine self-assessment accuracy among upper-secondary students, but this work has focused on LLM-generated feedback during writing rather than AI-assisted learning itself [8]. Meanwhile, experimental studies directly examining the relationship between AI-assisted performance and metacognitive accuracy have predominantly used adult populations and tasks such as logical reasoning or decision-making [2], [3], [4], [7]. Consequently, there remains limited experimental evidence on whether generative AI assistance during learning changes the accuracy with which adolescents judge their own knowledge, particularly when metacognitive calibration is operationalized through the correspondence between item-level confidence and objectively measured correctness [2], [8].

3.RESEARCH QUESTION:

•Primary/Central Research Question

How do three learning conditions—no generative AI, generative AI assistance, and generative AI assistance with structured metacognitive reflection—affect adolescents' metacognitive calibration immediately after learning and seven days later?

• Research Objectives

-> To examine the effect of GenAI-assisted learning on adolescents' learning performance.
-> To compare metacognitive calibration across AI-assisted and non-AI-assisted learning.
-> To examine confidence–performance alignment across immediate and delayed assessments.
-> To examine whether the relationship between confidence and objective performance differs between immediate and delayed assessments.

• Hypothesis

-> H1: Adolescents in the AI-assisted learning condition will demonstrate poorer metacognitive calibration than adolescents in the no-AI control condition, as indicated by a lower correspondence between item-level confidence judgments and objectively assessed correctness.

-> H2: Adolescents in the AI-assisted learning condition with structured metacognitive reflection will demonstrate better metacognitive calibration than adolescents in the AI-assisted learning condition without structured reflection.

-> H3: The effect of learning condition on metacognitive calibration will vary across the immediate and seven-day delayed assessments.

•Null Hypotheses

-> $H0_1$: There will be no difference in metacognitive calibration between the AI-assisted and no-AI control conditions.

-> $H0_2$: There will be no difference in metacognitive calibration between the AI-assisted condition with structured reflection and the AI-assisted condition without structured reflection.

-> $H0_3$: There will be no interaction between learning condition and assessment time on metacognitive calibration.

4. METHODOLOGY:

(4.1) Research Design

The study will employ a quantitative, randomized, three-condition experimental design with repeated assessments. Adolescents will be randomly assigned to one of three learning conditions: no generative AI, generative AI assistance, or generative AI assistance with structured metacognitive reflection. Following the learning intervention, participants will complete an immediate assessment and a parallel-form delayed assessment seven days later. Metacognitive calibration will be examined by comparing participants' confidence judgments with their objectively assessed correctness.

(4.2) Participants

The study will recruit adolescent learners within the predefined age range. Participants will be eligible if they fall within the specified age range, can understand the study instructions, have access to the required device and internet connection, and provide the required consent/assent. Participants who do not meet the eligibility criteria, do not provide the required consent, or fail to complete the necessary stages of the study will be excluded according to the predefined exclusion criteria.
The adolescent population is selected because existing experimental research on GenAI in education remains predominantly concentrated among university students, while evidence involving secondary and upper-secondary learners is comparatively limited and heterogeneous [6], [5], [8], [11].

(4.3) Sampling and Randomization

Participants will be recruited through the predetermined recruitment channels and will subsequently be randomly allocated to one of the three

experimental conditions in equal proportions. Random allocation will reduce systematic differences between groups and ensure that each participant has an equal probability of entering any condition. The intended allocation is 57 participants per condition, resulting in a target recruitment sample of 171 participants.

(4.4) Sample Size and Power

A target sample of 171 participants (57 per condition) will be recruited. The sample-size planning assumes a significance level of α = .05, statistical power of .80, a medium expected effect size (f = .25), and a correlation of .50 between repeated measurements. The recruitment target incorporates an allowance of approximately 7–8% for attrition, particularly at the seven-day delayed assessment. The calculation is based on a repeated-measures approximation to the planned mixed-effects analysis.

(4.5) Variables

The **independent variable** is learning condition (no AI, AI assistance, AI + structured metacognitive reflection). The **primary outcome** is metacognitive calibration, defined as the correspondence between confidence and objective correctness. Secondary outcomes include accuracy, confidence, calibration error, Brier score, calibration bias, metacognitive discrimination, and retention. **Assessment time** (immediate vs. seven days) is the repeated-measures factor.

(4.6) Operational Definitions

-> ***GenAI assistance:*** Access to the designated GenAI system during learning under a standardized protocol.
-> ***Structured metacognitive reflection:*** Standardized prompts requiring evaluation of understanding, uncertainty, confidence, and AI-output reliability.
-> ***Confidence:*** Estimated probability that an answer is correct, rated 0–100%.
-> ***Accuracy:*** Whether an assessment response is objectively correct.
-> ***Metacognitive calibration:*** Correspondence between confidence and objective correctness across items.
-> ***Retention:*** Knowledge demonstrated on the seven-day delayed assessment.

(4.7) Learning Materials

Participants will complete a standardized learning intervention covering three cognitive domains: selective attention and inattentional blindness; memory formation, reconstruction, and false memories; and cognitive biases in judgment. Each domain will contribute 12 assessment items (4 easy, 4 medium, 4 difficult), yielding 36 items per assessment. Learning content, assessment coverage, and learning time will be standardized across conditions so that differences primarily reflect the experimental manipulation.

(4.8) Experimental Conditions

-> **Condition 1:** No Generative AI.
Participants will complete the standardized learning intervention without access to a generative AI system or structured metacognitive-reflection prompts.

-> **Condition 2:** Generative AI Assistance.
Participants will complete the same learning intervention with access to the designated GenAI system under the standardized AI protocol. They will not receive the structured reflection intervention.

-> **Condition 3:** Generative AI Assistance + Structured Reflection.
Participants will receive the same GenAI access as Condition 2 and will additionally complete six standardized metacognitive-reflection prompts during learning.
All three conditions will receive comparable learning content and learning opportunities.

(4.9) AI Protocol

A single designated GenAI tool/model will be used consistently across both AI conditions, with standardized instructions defining permitted and prohibited uses. Participants may use AI for protocol-defined assistance, such as explanations or clarification, but not during assessments. Where feasible, AI interactions will be logged to monitor adherence and characterize use. Standardization is necessary given the variation in GenAI interaction across prior studies [5], [6].

(4.10) Reflection Protocol

Participants in the AI + structured reflection condition will complete six standardized reflection prompts, with two prompts following each learning domain.

• The prompts will target:
-> evaluation of AI-generated information;
-> recognition of uncertainty;
-> independent understanding;

-> verification of information;
-> confidence monitoring; and
-> evaluation of one's own knowledge.

The same prompts will be presented at the same stages of the learning intervention for all participants in this condition. The reflection manipulation is informed by research indicating that metacognitive reflection and verification can influence learners' interaction with AI-generated information [7], [11], [13].

(4.11) Experimental Procedure
After consent/assent, participants will complete the baseline stage and be randomly assigned to one of three learning conditions. They will then complete the standardized intervention according to their assigned condition, with the two AI conditions receiving access to the designated GenAI system and one additionally completing structured reflection prompts. Immediately after learning, all participants will complete an AI-free assessment, followed by a parallel-form AI-free assessment seven days later.

•The procedure can therefore be summarized as:
Consent/assent → Baseline → Randomization → Learning intervention → Condition-specific manipulation → Immediate AI-free assessment → 7-day interval → Delayed AI-free assessment.

(4.12) Assessment Design
The immediate AI-free assessment will contain 36 items: 12 each from selective attention, memory, and cognitive biases, with four items at each difficulty level per domain. For each item, participants will provide an answer followed by a 0–100% confidence rating. After seven days, they will complete a 36-item parallel-form assessment covering the same domains with comparable difficulty and distribution. Both assessments will be completed without GenAI assistance, enabling independent measurement of performance and confidence.

(4.13) Measurement of Calibration
Metacognitive calibration will be assessed through the correspondence between item-level confidence (0–100%) and correctness coded as 1 for a correct response and 0 for an incorrect response. The primary measure, Absolute Calibration Error (ACE), will capture the absolute discrepancy between confidence and observed correctness, with lower values indicating better calibration. Brier score, calibration bias, and metacognitive discrimination will provide complementary measures of accuracy, directional bias, and confidence–performance discrimination. This item-level approach distinguishes calibration from broader measures of metacognitive awareness used in previous GenAI research [2], [10], [14].

(4.14) Data Quality and Exclusion Criteria
Data-quality and exclusion criteria will be established before examination of the outcome results. These will include predefined requirements for study completion, handling of missing responses, invalid or technically compromised sessions, delayed-assessment attrition, duplicate participation, and other specified protocol violations.
The same predefined criteria will be applied consistently across experimental conditions. Participants will not be excluded based on whether their results support or contradict the study hypotheses.

(4.15) Data Analysis
The **primary analysis** will use the planned mixed-effects analytical framework to examine the effects of learning condition, assessment time, and their interaction on metacognitive calibration.
The **principal analysis** will examine whether calibration differs among the three learning conditions and whether these differences change between the immediate and seven-day delayed assessments.
Secondary analyses will examine differences in knowledge accuracy, confidence, retention, calibration error, Brier score, calibration bias, and metacognitive discrimination.

-> The analysis will therefore distinguish the effect of GenAI assistance from the additional effect associated with structured metacognitive reflection.

(4.16) Ethical Considerations
As the study involves adolescent participants, appropriate ethical safeguards will be implemented. Participation will require the applicable informed consent/parental consent and participant assent. Participation will be voluntary, and participants will have the right to withdraw according to the study protocol.
Participant information will be kept confidential and, where possible, pseudonymized. Data will be securely stored and used only for the stated research purposes. The study will minimize the collection of personally identifiable information and will implement appropriate safeguards for adolescent participants and their digital data.

5. RESEARCH TOPIC:

(5.1) Why This Topic Is Timely

The rapid integration of Generative Artificial Intelligence into learning environments has created new forms of interactive and personalized educational assistance. Existing research indicates that GenAI can influence academic performance, confidence, reliance, and cognitive processes, while its effects on metacognitive accuracy remain less consistent [2], [4], [5], [7], [8]. This issue is particularly timely for adolescent learners, as GenAI is becoming increasingly accessible while experimental evidence concerning its effects on their metacognitive monitoring remains comparatively limited [6], [8], [10], [11].

(5.2) Innovation And Creative Problem-Solving

The proposed study addresses a specific problem in AI-assisted learning: improved performance with GenAI may not necessarily correspond to accurate understanding of what has been learned [2], [4], [8]. Rather than examining this discrepancy only as an outcome, the study experimentally tests a potential solution through structured metacognitive reflection [7], [13]. Participants in the AI + structured reflection condition will receive standardized prompts targeting uncertainty, verification of AI-generated information, independent understanding, confidence monitoring, and evaluation of their own knowledge.

The three-condition design—no AI, AI assistance, and AI assistance with structured reflection—will allow the study to distinguish the effect of AI assistance from the potential contribution of reflection. Item-level confidence ratings paired with objective correctness will then determine whether the reflection intervention is associated with improved confidence–performance calibration, both immediately and seven days later.

6. EXPECTED OUTCOME/CONTRIBUTION:

This study will examine whether GenAI-assisted learning affects adolescents' metacognitive calibration and whether these effects persist after seven days. By comparing no AI, AI assistance, and AI assistance with structured metacognitive reflection, the study will distinguish the effect of AI assistance from the potential contribution of reflection. Item-level confidence ratings paired with objectively measured correctness will further examine whether structured reflection is associated with improved confidence–performance calibration, both immediately and after seven days. Findings may contribute to research on adolescent metacognition and inform AI-supported learning designs that incorporate independent evaluation, verification, and reflection.

7. LIMITATIONS:

Several limitations should be considered when interpreting the findings. First, the adolescent sample and recruitment setting may limit generalizability across age groups, educational systems, and learning contexts, particularly given the predominance of university-based GenAI research [5], [6]. Second, the controlled setting and standardized materials may not reflect the complexity of everyday GenAI use across tasks and domains [2], [8]. Third, prior GenAI familiarity, the selected AI model and protocol, and seven-day attrition may influence interaction patterns and longitudinal comparisons. Finally, confidence–performance calibration captures only one aspect of metacognition and may not reflect broader awareness, regulation, or beliefs [10], [12], [14]. Although parallel forms will be matched for content and difficulty, exact equivalence cannot be assured.

8. PROJECT PRACTICALITIES/FEASIBILITY:

The study is feasible in a controlled digital environment with 171 adolescents (57 per condition) randomly assigned to three conditions. Standardized digital procedures will deliver the intervention and AI-free assessments immediately and after seven days. Attrition will be minimized through reminders, monitored across conditions, and transparently reported, with available follow-up data retained where appropriate.

9. CONCLUSION:

As GenAI becomes embedded in learning, it is important to examine not only performance with AI assistance but also learners' ability to accurately assess their own knowledge. This study addresses this gap in adolescents through a three-condition design comparing no AI, AI assistance, and AI assistance with structured metacognitive reflection. Confidence–accuracy calibration will be measured immediately and after seven days, distinguishing objective performance from self-assessment to clarify GenAI's relationship with adolescents' metacognitive monitoring.

10. POST-PROGRAM/DISSEMINATION PLAN:

Findings could be disseminated through research reports or presentations to secondary-school students, parents/guardians, teachers, school counsellors, and school administrators, as well as educational researchers and AI-in-education practitioners. Dissemination would highlight GenAI-assisted learning, performance, and metacognitive calibration, with practical emphasis on independent assessment, verification, and reflection. Findings would be reported objectively and interpreted within the study's population and experimental conditions.

REFERENCES

[1] M. Alier, F. J. García-Peñalvo, and J. D. Camba, "Generative artificial intelligence in education: From deceptive to disruptive," *International Journal of Interactive Multimedia and Artificial Intelligence*, vol. 8, no. 5, pp. 5–14, 2024, doi: 10.9781/IJIMAI.2024.02.011.

[2] D. Fernandes, S. Villa, S. Nicholls, O. Haavisto, D. Buschek, A. Schmidt, T. Kosch, C. Shen, and R. Welsch, "AI makes you smarter but none the wiser: The disconnect between performance and metacognition," *Computers in Human Behavior*, vol. 175, Art. no. 108779, 2026, doi: 10.1016/j.chb.2025.108779.

[3] F. Tang, F. Yao, and M. Zhou, "Innovating diagnostic learning: How generative AI explanation styles influence cognitive load and confidence calibration in medical students," *Computers & Education*, vol. 249, Art. no. 105619, 2026, doi: 10.1016/j.compedu.2026.105619.

[4] M. Urban, F. Děchtěrenko, J. Lukavský, V. Hrabalová, F. Svacha, C. Brom, and K. Urban, "ChatGPT improves creative problem-solving performance in university students: An experimental study," *Computers & Education*, vol. 215, Art. no. 105031, 2024, doi: 10.1016/j.compedu.2024.105031.

[5] S. Chen and A. C. K. Cheung, "Effect of generative artificial intelligence on university students' learning outcomes: A systematic review and meta-analysis," *Educational Research Review*, vol. 49, Art. no. 100737, 2025, doi: 10.1016/j.edurev.2025.100737.

[6] R. Deng, M. Jiang, X. Yu, Y. Lu, and S. Liu, "Does ChatGPT enhance student learning? A systematic review and meta-analysis of experimental studies," *Computers & Education*, vol. 227, Art. no. 105224, 2025, doi: 10.1016/j.compedu.2024.105224.

[7] S. R. Ren, "College students' metacognitive awareness of generative-AI reliance: An experimental study of decision confidence and attribution bias," *Frontiers in Psychology*, vol. 17, Art. no. 1926110, 2026, doi: 10.3389/fpsyg.2026.1926110.

[8] L. W. Liebenow, F. T. C. Schmidt, J. Meyer, and J. Fleckenstein, "Self-assessment accuracy in the age of artificial intelligence: Differential effects of LLM-generated feedback," *Computers & Education*, vol. 237, Art. no. 105385, 2025, doi: 10.1016/j.compedu.2025.105385.

[9] M. Urban, C. Brom, J. Lukavský, F. Děchtěrenko, V. Hein, F. Svacha, P. Kmoníčková, and K. Urban, "'ChatGPT can make mistakes. Check important info.' Epistemic beliefs and metacognitive accuracy in students' integration of ChatGPT content into academic writing," *British Journal of Educational Technology*, vol. 56, no. 5, pp. 1897–1918, 2025, doi: 10.1111/bjet.13591.

[10] K. W. Yau, T. Dong, C. S. Chai, T. K. F. Chiu, H. Meng, I. King, S. W. H. Wong, and Y. Yam, "An epistemic network analysis (ENA) of secondary school students' metacognition in learning generative AI," in *Proc. 2025 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)*, 2025, doi: 10.1109/TALE66047.2025.11346635.

[11] C. Zhou, "From initial trust to critical reconstruction: Upper secondary students' engagement with generative AI in science learning," *Frontiers in Psychology*, vol. 17, Art. no. 1830646, 2026, doi: 10.3389/fpsyg.2026.1830646.

[12] J. Iqbal, Z. F. Hashmi, M. Z. Asghar, and M. N. Abid, "Generative AI tool use enhances academic achievement in sustainable education through shared metacognition and cognitive offloading among preservice teachers," *Scientific Reports*, vol. 15, Art. no. 16610, 2025, doi: 10.1038/s41598-025-01676-x.

[13] J. Liu, Y. Zhang, W. Li, Q. Wang, P. Niu, and X. Zhang, "Adaptive vs. planned metacognitive scaffolding for computational thinking: Evidence from generative AI-supported programming in elementary education," *Computers & Education*, vol. 241, Art. no. 105473, 2026, doi: 10.1016/j.compedu.2025.105473.

[14] Y. Xie, Y. Li, T. Yu, and Y. Liu, "Metacognitions about generative AI use: Psychometric and network analysis among Chinese college students," *Education and Information Technologies*, vol. 30, pp. 20523–20542, 2025, doi: 10.1007/s10639-025-13584-8.

[15] M. Chavan and R. Patole, "Generative AI as a catalyst for metacognitive growth and academic performance: A quantitative study in secondary education," *International Journal of Advanced Trends in Engineering and Management*, vol. IV, no. 8, pp. 26–38, 2025, doi: 10.59544/CBKU8444/IJATEMV04I08P3.

[16] A. Barcaui, "ChatGPT as a cognitive crutch: Evidence from a randomized controlled trial on knowledge retention," *Social Sciences & Humanities Open*, vol. 12, Art. no. 102287, 2025, doi: 10.1016/j.ssaho.2025.102287.